

In most stars, nebulae, H II regions, and other astronomical sources, hydrogen and helium are the two dominant elements. The hydrogen mass fraction is generally expressed as $X \equiv \frac{m_{\text{H}}}{M}$, where M is the total mass of the system, and m_{H} is the mass of the hydrogen it contains. Similarly, the helium mass fraction is denoted as $Y \equiv \frac{m_{\text{He}}}{M}$. The remainder of the elements are collectively referred to as "metals", and the mass fraction of metals is calculated as

$$Z = \sum_{e>\text{He}} \frac{m_e}{M} = 1 - X - Y .$$

For the surface of the Sun (symbol ☉), these parameters are measured to have the following values:^[16]

Description	Solar value
Hydrogen mass fraction	$X_{\odot} = 0.7381$
Helium mass fraction	$Y_{\odot} = 0.2485$
Metal mass fraction	$Z_{\odot} = 0.0134$

Due to the effects of stellar evolution, neither the initial composition nor the present day bulk composition of the Sun is the same as its present-day surface composition.

Chemical abundance ratios

The overall stellar metallicity is conventionally defined using the total hydrogen content, since its abundance is considered to be relatively constant in the Universe, or the iron content of the star, which has an abundance that is generally linearly increasing in time in the Universe.^[17] Hence, iron can be used as a chronological indicator of nucleosynthesis. Iron is relatively easy to measure with spectral observations in the star's spectrum given the large number of iron lines in the star's spectra^[18] (even though oxygen is the most abundant heavy element – see metallicities in H II regions below). The abundance ratio is the common logarithm of the ratio of a star's iron abundance compared to that of the Sun and is calculated thus:^[19]

$$\left[\frac{\text{Fe}}{\text{H}} \right] = \log_{10} \left(\frac{N_{\text{Fe}}}{N_{\text{H}}} \right)_{\star} - \log_{10} \left(\frac{N_{\text{Fe}}}{N_{\text{H}}} \right)_{\odot} ,$$

where N_{Fe} and N_{H} are the number of iron and hydrogen atoms per unit of volume respectively, ☉ is the standard symbol for the Sun, and $\star$ for a star (often omitted below). The unit often used for metallicity is the dex, contraction of "decimal exponent".^[20] By this formulation, stars with a higher metallicity than the Sun have a positive common logarithm, whereas those more dominated by hydrogen have a corresponding negative value. For example, stars with a $\left[\frac{\text{Fe}}{\text{H}} \right]_{\star}$ value of +1 have 10 times the metallicity of the Sun (10^{+1}); conversely, those with a $\left[\frac{\text{Fe}}{\text{H}} \right]_{\star}$ value of −1 have $\frac{1}{10}$, while those with a $\left[\frac{\text{Fe}}{\text{H}} \right]_{\star}$ value of 0 have the same metallicity as the Sun, and so on.^[21]